Anmol Soni

PhD Candidate in Computational Climate Economics

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Research Profile

Computational climate economist investigating distributional economic and behavioural impacts of extreme weather events and climate policy. Research methods combine SFC agent-based macroeconomic modelling with survey data and natural language processing.

Research Interests

Distributional impacts of climate change and climate policy; regional inequality, survey design analysis, computational models and text-based empirical methods.

Education

PhD Candidate, Computational Climate Economics **Feb 2023–Jan 2027**
School of Business and Economics, Maastricht University, Netherlands

Expected completion: January 2027. Research on extreme weather events, climate narratives, heterogeneous regional economic effects and climate-policy responses, using computational and empirical approaches.

MSc, Engineering and Policy Analysis (Honours) **Aug 2020–Aug 2022**
Delft University of Technology, Netherlands

Focus on complex adaptive socio-technical systems, modelling and simulation, policy analysis and data analytics. Thesis: *Exploring demand response opportunities in energy communities*.

Postgraduate Diploma Traineeship, Power Engineering **2015–2016**
National Power Training Institute, Neyveli, India

Bachelor of Mechanical Engineering **Jul 2011–Jul 2015**
Rashtrasant Tukadoji Maharaj Nagpur University, India

Graduated with First Division; specialisation in Energy & Power Plant Engineering.

Publications

Working Papers and Research Manuscripts

- **Soni, A.**, Treibich, T., Comes, T. and Piccillo, G. (2026). *Narrating the Deluge: Media and residents diverge in subjective attribution roles following flooding.* LEM Working Paper Series, No. 2026/13, Sant'Anna School of Advanced Studies. DOI: 10.57838/sssa/8hzz-6k30.
- **Soni, A.**, Comes, T., Lamperti, F., Piccillo, G., Roventini, A. and Treibich, T. (2026). *Regional climate shocks, adaptation policy and distributional outcomes in an agent-based model.* Working paper in preparation.

Policy and Research Reports

1. Hoppe, T., Fremouw, M. A., **Soni, A.**, Manktelow, C. and Bickerstaff, K. (2023). *SHIFFT WP2 Monitoring & Evaluation of the Co-creation Pilots: WP2 (Work Package-2) Monitoring and Evaluation Report.* SHIFFT. Policy report.

Research Experience

PhD Researcher, Computational Climate Economics **Feb 2023–Jan 2027**
Maastricht University, School of Business and Economics, Netherlands

- Investigate how climate risks and policy responses produce heterogeneous macroeconomic, regional and behavioural outcomes.
- Analyse systematic newspaper text data and a representative post-event survey following the 2023 Emilia-Romagna floods; use natural language processing and regression analysis to examine attribution of responsibility, differences in flood exposure and self-reported mitigation behaviour.
- Extend the DSK stock-flow-consistent agent-based macroeconomic model to multiple regions, incorporating regional climate shocks, adaptation policies and an opinion-dynamics component to explore distributional outcomes and macroeconomic trajectories.
- Collaborate on interdisciplinary research with Maastricht University, TU Delft and Sant'Anna School of Advanced Studies; communicate findings at research seminars and international conferences.

Visiting Positions

Visiting Doctoral Researcher **Jun 2026–Aug 2026**
Institute of Economics (LEM), Sant'Anna School of Advanced Studies, Pisa, Italy

- Worked on the multi-region agent-based macroeconomic (DSK) modelling project examining heterogeneous regional climate shocks, adaptation policies and distributional outcomes.
- Presented research at a Sant'Anna seminar and engaged with researchers working on evolutionary and computational economics.

Research Assistantships

Research Assistant, Energy Transition Monitoring and Evaluation **Oct 2022–Dec 2022**
TU Delft, Department of Multi Actor Systems, Netherlands | Interreg 2 Seas SHIFFT project

- Co-authored the monitoring and evaluation report on sustainable-heating co-creation across six pilots in Belgium, the Netherlands, France and the United Kingdom.
- Contributed to monitoring methodology, evaluation of project pilots and a calculation tool estimating pilot-level CO_2 impacts.

Research Assistant, Spatial (GIS) Analysis **Apr 2021–Feb 2022**
TU Delft, TPM Resilience Lab, Netherlands | EU HERoS project

- Curated geographical location data and developed GIS module for an agent-based simulation / digital-twin model of The Hague in research on urban resilience and crisis governance.
- Presented research findings at the International Conference on Computational Social Science (2021).

Additional Research Outputs and Projects

- **MSc thesis: Exploring demand response opportunities in energy communities (2022).** Developed an agent-based model to assess and optimise household and non-residential demand response, behind-the-meter storage and electricity-grid dependence. Thesis undertaken in collaboration with Croonwolter&dros, Rotterdam (Feb–Aug 2022).
- **Geothermal integration in Rotterdam's Waalhaven area (2021–2022).** Interdisciplinary TU Delft project in collaboration with the Municipality of Rotterdam. Contributed an actor-network scan, social-engagement strategy and heating-demand estimates to a roadmap for integrating geothermal energy into urban planning; coordinated a three-student team.
- **QCONNECT: Circularity in the Dutch automotive sector (2021–2022).** Developed a material-flow agent-based model for exploratory analysis of circular-economy policy options in a TU Delft project associated with RIVM.

Selected Conference and Research Seminar Presentations

Regional climate shocks, adaptation policy and distributional outcomes in an agent-based model

- Computing in Economics and Finance (CEF), Venice, Italy, June 2026.
- Workshop on Economic Heterogeneous Interacting Agents (WEHIA), Ancona, Italy, June 2026.
- Institute of Economics / Sant'Anna research seminar, Pisa, Italy, June 2026.
- MILE Seminar, Maastricht University, Netherlands, May 2026.
- Workshop on Macroeconomic Implications of Climate-Related Risks, Pisa, February 2026.

Narrating the Deluge: Media and residents diverge in subjective attribution roles following flooding

- International Climate Resilience Conference (iCARE), LMU Munich, Germany, October 2025.
- Workshop on Political and Legal Opportunities and Obstacles to Climate Action in Italy, Johns Hopkins SAIS Europe, Bologna, April 2024.
- MORSE Conference, Maastricht University, Netherlands, October 2023.

Earlier research presentation

- EU HERoS spatial modelling research, International Conference on Computational Social Science, 2021.
- Systems-approach for optimisation of flux-production, International Conference on Emerging Trends in Engineering and Technology, Nagpur, India, 2014.

Teaching and Thesis Supervision

Tutor and Bachelor's Thesis Supervisor **2022–2026**
School of Business and Economics, Maastricht University, Netherlands

- **Sustainable Development**, BSc Business Engineering (2023–2026).
- **Emerging Markets in the Global Economy**, BSc International Business (2023–2026).
- **International Economic Relations**, BSc Economics and Business Economics (2024–2025).
- **Economy Game**, undergraduate market simulations (2023–2024).
- **Presentation Skills**, undergraduate academic communication (2022–2023).
- Supervised bachelor's theses on economic and emerging-market topics, including foreign direct investment and industrial development, real estate markets, and drought impacts in Sub-Saharan Africa.

Grants and Research Funding

- **International Research Travel Grant**, Graduate School of Business and Economics (GSBE), Maastricht University (May 2026).
- **Primary Data Collection Grant**, Graduate School of Business and Economics (GSBE), Maastricht University (August 2023).

Academic Service and Professional Memberships

- **Journal Reviewer**, *Journal of Economic Dynamics and Control* (October 2026).
- **Seminar Organiser**, MILE Seminar Organising Committee, School of Business and Economics, Maastricht University (January 2026–present).
- **Member**, Society for Computational Economics (February 2024–present).

Leadership and Institutional Representation

- **Faculty Ambassador**, TU Delft Global Student Club (December 2021–August 2022).
- **Treasurer**, Study Association of Engineering and Policy Analysis, TU Delft (November 2020–December 2021).

Previous Policy and Energy-Sector Experience

Research Associate, Energy Policy **Jun 2018–Jul 2020**
Centre for Energy, Environment & People, Rajasthan, India

- Worked on electricity governance and consumer capacity-building initiatives, helping consumers participate in tariff-setting and regulatory processes.
- Conducted field visits to document electricity consumers' grievances and barriers to redress in rural Rajasthan; contributed to energy-policy and regulatory engagement projects.

Design Engineer, Solar PV **Dec 2017–May 2018**
Vibgyor Energy, India

- Worked on the design and commissioning of more than 30 MW of solar PV capacity using PVsyst and SketchUp; coordinated two drafting staff and three survey engineers.

Design Engineer, Solar Microgrids **Jul 2017–Nov 2017**
Boond, India

- Engineered rural solar-microgrid electrification projects serving approximately 300 households in Uttar Pradesh and Rajasthan.

Graduate Engineering Trainee **Dec 2016–Jun 2017**
Boond, India

Industrial Trainee **Jun 2013**
NTPC, Mouda, India

Research Methods and Technical Skills

- **Empirical methods:** Survey data, regression analysis, natural language processing, systematic text analysis, data gathering, web scraping and interpretation of heterogeneous responses.
- **Computational economics:** Agent-based modelling, stock-flow-consistent macroeconomic models, regional simulation, model evaluation and policy-scenario analysis.
- **Programming and research software:** Python, R, C++; Pandas, Matplotlib, Plotly, ggplot2; Mesa, NetLogo, LSD; Git and GitHub.
- **Additional methods:** GIS / spatial analysis; multi-criteria decision-making; uncertainty analysis; socio-technical energy-system modelling.

Selected Training and Short Courses

- **Agent-Based Modelling in Social Sciences**, UNU-MERIT, Netherlands (September 2025).
- **Agent-Based Models in Economics**, Sant'Anna School of Advanced Studies, Italy (July 2024).
- **Intangible Matter of Spaces**, Alta Scuola Politecnica Winter School, Italy (March 2022).
- **Energy Economics and Policy**, Indian Institute of Technology (November 2019).
- **Python for Data Science**, Indian Institute of Technology (November 2019).

Languages

Hindi (native); English (native); Dutch (vantage).

Academic References

Tina Comes
Full Professor

Director, Resilience Lab & Domain Lead Disasters, Faculty of Technology, Policy and Management, TU Delft

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Associate Professor

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